

MMV: A Lifecycle-Based Ontology for Declarative Mapping projects

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Abstract— Declarative mappings support semantic interoperability across heterogeneous data sources, enabling tasks such as ontology alignment, data uplift, and interlinking. However, mapping projects are often documented only through their final artefacts, with limited structured metadata describing their context, development process, and evolution. This paper presents MMV (Mapping Metadata Vocabulary), a lifecycle-based ontology designed to represent metadata for declarative mapping projects across different mapping types. MMV captures contextual and procedural metadata spanning analysis, design, development, testing, and maintenance, while reusing established Semantic Web vocabularies. The paper describes the design of MMV, illustrates its instantiation, and outlines a planned expert evaluation.

Keywords— *declarative mappings, mapping metadata, ontology design, mapping lifecycle, semantic interoperability*

I. INTRODUCTION

Declarative mappings play a central role in enabling semantic interoperability across heterogeneous data sources, supporting tasks such as data transformation, ontology alignment, and interlinking [1]. In this work, a *mapping* is understood as a declarative specification that defines correspondences or transformations between heterogeneous data models, schemas, or ontologies. Such mappings are typically expressed using formal rule-based languages and are executed or evaluated by mapping engines as part of knowledge graph construction, data integration, and semantic evaluation workflows.[1]

Despite their importance, mapping projects are commonly documented primarily through their final artefacts (e.g., mapping files or alignment outputs), with limited structured metadata describing how and why those artefacts were produced [2]. Contextual information such as mapping purpose, domain assumptions, tool configurations, validation procedures, quality measurements, and maintenance history is often scattered across informal documentation or omitted entirely. In addition, mappings increasingly are complex spanning the entire range of complex correspondences patterns [3] and incorporating functions for transformation logic [4]. This lack of systematic metadata hinders traceability, complicates comparison across projects, and limits the reliable reuse of mapping artefacts in new contexts.

These challenges are particularly evident in large-scale and collaborative initiatives, where mappings are developed, evaluated, and revised by multiple stakeholders over time. Without lifecycle-oriented metadata, it becomes difficult to analyze how mappings evolve across versions and to support transparency and governance.

This paper builds on a body of our prior work that has examined mapping metadata from complementary perspectives. Previous studies investigated (i) the metadata needs of mapping practitioners through a community survey, (ii) the design and implementation of MetaSEMAP, a web-based tool for capturing mapping metadata, (iii) the usability of structured metadata annotation through usability user studies, and (iv) the role of large language models (LLMs) in generating declarative mappings with and without structured metadata guidance [5],[6],[7],[8]. Collectively, these works indicate that lifecycle-oriented metadata is both needed and practically capturable, but also highlights the absence of a clearly articulated, reusable ontology to structure such metadata consistently across mapping types and tools.

In this paper, we address this gap by describing the Mapping Metadata Vocabulary (MMV) as a formal ontology. MMV is a lifecycle-based ontology designed to represent metadata about declarative mapping projects across analysis, design, development, testing, and maintenance. MMV is independent of specific mapping languages or execution frameworks and is intended to support a range of declarative mapping types, including ontology alignment, uplift mappings, and interlinking. MMV is instantiated through **MetaSEMAP** [6,7], which provides a structured interface for creating MMV-compliant metadata without requiring users to directly model OWL.

To assess the adequacy of MMV as an ontology, in this paper we outline a planned expert evaluation focusing on conceptual clarity, coverage, and alignment with established ontology engineering best practices. By formalising lifecycle-oriented metadata in a reusable ontology, this work aims to support more systematic documentation, analysis, and reuse of declarative mapping projects.

II. MOTIVATING MAPPING PROJECTS AND METADATA NEEDS

A. Motivating Example

As an illustrative case, we use the OAEI 2025 Anatomy track to exemplify one instance of the broader challenges associated with documenting declarative mapping projects.

The Ontology Alignment Evaluation Initiative (OAEI) has organized annual evaluation campaigns since 2004 to assess ontology matching systems across multiple tracks, datasets, and evaluation settings [9]. For each track, participating systems generate alignment artefacts that are published as independent RDF alignment files, together with aggregated evaluation results based on standard metrics such as precision, recall, F-measure, recall+, and coherence.

In the OAEI 2025 campaign [10] Anatomy track for example, systems generate alignments between the Adult Mouse Anatomy ontology and the NCI Thesaurus. The resulting raw alignment files are made publicly available per system and per track, while details about the experimental setting such as the evaluation platform used (e.g., MELT, SEALS, or developer-provided execution), configuration assumptions, coherence checks, and known limitations are described separately in accompanying documentation [11].

Although this setup enables quantitative comparison of systems within the campaign, the alignment artefacts themselves are published without structured, machine-readable metadata describing how they were produced, under which assumptions, or how they relate to other artefacts across executions or versions. As a result, it is difficult to systematically trace the provenance of individual alignments, compare artefacts produced under different conditions, or assess their suitability for reuse beyond the original evaluation context.

This situation reflects a broader challenge in declarative mapping projects: mapping artefacts are commonly shared as standalone outputs, detached from lifecycle-oriented metadata describing their creation, testing, and evolution. Addressing this gap motivates the need for a structured metadata vocabulary for declarative mapping projects.

B. Metadata Requirements Across the Lifecycle

The motivating example illustrates that declarative mapping projects encompass more than the production of final mapping artefacts. Across their lifecycle, such projects involve a series of activities ranging from initial analysis to long-term maintenance each of which introduces contextual factors that influence how mappings are interpreted, developed and evaluated.

At early stages, metadata is required to describe the purpose and scope of a mapping project, including intended use, domain assumptions, and requirements that guide modelling decisions. During design and development, metadata is needed to document implementation choices, such as selected approaches, tools, configurations, and dependencies. Testing and evaluation phases introduce the need for metadata describing validation activities, including evaluation criteria, quality metrics, and assessment outcomes. Finally, maintenance phases require metadata that captures versioning, derivation, updates, and responsible stakeholders, enabling traceability across previous mapping versions.

In practice, this information is often fragmented across informal documentation or omitted entirely. The absence of a unified, lifecycle-oriented metadata model makes it difficult to systematically compare mapping projects, trace how mapping artefacts evolve, or assess their suitability for reuse across settings.

Based on these observations, we identify the following requirements for mapping metadata:

- R1 (Project-level context): Metadata must describe mapping projects as first-class entities, including purpose, scope, stakeholders, and temporal boundaries.
- R2 (Lifecycle activities): Metadata must capture distinct lifecycle activities (analysis, design, development, testing, maintenance) and their relationships to mapping artefacts.
- R3 (Artefact documentation): Metadata must describe mapping artefacts, including type, language, version, and derivation relationships.

These requirements motivate the design of a lifecycle-based ontology for mapping metadata, which we introduce in the following section.

III. MMV ONTOLOGY DESIGN AND STRUCTURE

This section presents MMV (Mapping Metadata Vocabulary), a lifecycle-based ontology for representing metadata about declarative mapping projects. MMV is designed to capture not only mapping artefacts, but also the activities, assumptions, evaluation procedures, and maintenance actions that shape their creation. The ontology is intended to support documentation, traceability, and reuse across different declarative mapping types, including uplift mappings, ontology alignment, and interlinking.

A. Ontology Engineering Approach and Scope

The development of MMV followed established ontology engineering practices, drawing in particular on principles from lightweight ontology development methodologies such as the Linked Open Terms (LOT) approach [12]. These methodologies emphasize requirement-driven design, reuse of existing vocabularies, and iterative refinement informed by evaluation. MMV adopts these principles to ensure conceptual clarity, modularity, and interoperability.

MMV is scoped to the project level, modelling mapping as evolving processes rather than isolated files. The ontology does not prescribe a specific mapping language or execution framework, allowing it to be applied uniformly across different declarative mapping scenarios.

B. Competency Questions and Requirements

The design of MMV was guided by a set of competency questions derived from observed metadata gaps in real-world mapping projects and evaluation campaigns. These questions capture the types of queries the ontology is intended to support and informed the identification of core concepts and relationships. Table 1 below summarizes the relationship

between these competency questions and the main ontology elements they motivated.

Table 1 Competency questions and Corresponding MMV elements.

CQ	Competency Question	Relevant MMV classes and properties
CQ1	What is the purpose and scope of a mapping project?	<i>mmv:MappingProject</i> , <i>mmv:hasAnalysis</i> , <i>mmv:hasRequirement</i> , <i>mmv:hasAssumption</i>
CQ2	Which domain and resources does the mapping operate on	<i>mmv:MappingProject</i> , <i>mmv:DataSet</i> , <i>prov:used</i>
CQ3	Which mapping artefacts were produced and in which format?	<i>mmv:MappingArtifact</i> , <i>mmv:mappingType</i> , <i>mmv:mappingLanguage</i> , <i>mmv:version</i>
CQ4	Which lifecycle activities led to the creation of the mapping?	<i>mmv:AnalysisActivity</i> , <i>mmv:DesignActivity</i> , <i>mmv:DevelopmentActivity</i>
CQ5	How was the mapping tested or validated	<i>mmv:TestingActivity</i> , <i>mmv:testsArtifact</i>
CQ6	How was the quality of the mapping assessed?	<i>mmv:EvaluationActivity</i> , <i>mmv:evaluatesArtifact</i> , <i>dqv:QualityMeasurement</i>
CQ7	Which tools were used during development and evaluation?	<i>mqv:MappingTool</i> , <i>prov:used</i>
CQ8	How has the mapping evolved over time?	<i>mmv:MaintenanceActivity</i> , <i>mmv:version</i> , <i>prov:wasDerivedFrom</i>
CQ9	Is the mapping suitable for reuse in a different context?	<i>mmv:MappingProject</i> , lifecycle activities, quality metadata

C. Core Concepts

At the centre of MMV is the concept of *MappingProject*, representing a coherent unit of work aimed at achieving a specific mapping objective. A *MappingProject* may involve one or more *MappingArtifact* instances, such as RML files, ontology alignments, or interlinking specifications.

MMV explicitly distinguishes between projects, activities, and artefacts. This modelling choice enables the association of multiple artefacts with shared context, including involved agents, tools, assumptions, temporal boundaries, and evaluation evidence.

D. Lifecycle Modeling

To represent the evolution of mapping projects, MMV models lifecycle phases as subclasses of *prov:Activity*. The ontology distinguishes between *AnalysisActivity*, *DesignActivity*, *DevelopmentActivity*, *TestingActivity*, and *MaintenanceActivity*, each capturing metadata relevant to that stage.

Early lifecycle activities document purpose, assumptions, requirements, and risks. Development activities capture implementation details such as mapping language and tools, while testing and evaluation activities record validation

procedures and quality assessments. Maintenance activities represent updates and derivation relationships between mapping versions. Table 2 below summarises the lifecycle phases and the types of metadata captured.

Table 2 Lifecycle phases and metadata captured in MMV

Lifecycle Phase	MMV class	Main MMV Properties/Metadata
Analysis	<i>mmv:AnalysisActivity</i>	<i>mmv:hasRequirement</i> , <i>mmv:hasAssumption</i> , <i>mmv:hasRiskOrIssue</i>
Design	<i>mmv:DesignActivity</i>	Design decisions and reuse considerations (linked via <i>mmv:hasDesign</i>)
Development	<i>mmv:DevelopmentActivity</i>	<i>mmv:MappingArtifact</i> , <i>mmv:mappingType</i> , <i>mmv:mappingLanguage</i> , <i>mmv:version</i> , <i>mqv:MappingTool</i>
Testing/Evaluation	<i>mmv:TestingActivity</i> <i>mmv:EvaluationActivity</i>	Validation activities linked via <i>mmv:testsArtifact</i> <i>mmv:evaluatesArtifact</i> , <i>dqv:Metric</i> , <i>dqv:QualityMeasurement</i>
Maintenance	<i>mmv:MaintenanceActivity</i>	Updates and evolution actions linked via <i>mmv:hasMaintenance</i>

E. Reuse of Existing Vocabulary

MMV follows a reuse-oriented design by aligning its core concepts with established Semantic Web vocabularies. *MappingProject* and *MappingArtifact* are modelled as *prov:Entity*, while lifecycle phases are modelled as *prov:Activity*, enabling provenance-aware descriptions. Quality-related metadata reuse concepts from DQV [13], and MQV [14] is used to represent mapping tools and assessment activities. This alignment ensures interoperability while allowing MMV to introduce domain-specific structure.

F. Ontology Overview and Use Cases

MMV structures metadata around explicit relationships between MappingProject, lifecycle activities, and resulting MappingArtifact instances. A MappingProject is linked to its lifecycle phases via properties such as hasAnalysis, hasDesign, hasDevelopment, hasTesting, and hasMaintenance, enabling traceability between activities, artefacts, and evaluation outcomes (Figure 1).

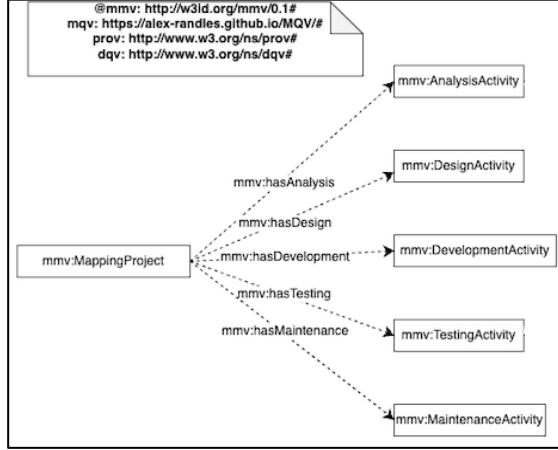


Figure 1 summarizes the main classes and relationships in MMV.

Figure 1 presents a partial view of the MMV ontology diagram, generated using the CHOWLK [15] ontology diagramming tool. Due to space limitations, the complete diagram is provided in the online ontology documentation and specification¹.

To illustrate the application of MMV across different declarative mapping types, the following use cases show partial instantiations of the ontology. Due to space limitations, each example shows only a subset of the metadata rather than a full lifecycle description.

- Use Case 1: Ontology Alignment (Analysis Phase)

This use case (Figure 2) illustrates how MMV captures analysis-phase metadata for an ontology alignment task. A MappingProject is linked to an AnalysisActivity that records a textual description of the task, key requirements, and identified risks associated with aligning mouse and human anatomy ontologies. The example focuses on documenting early analytical considerations and constraints that inform later design and implementation decisions.

```

1 @prefix mmv: <http://w3id.org/mmV/#> .
2 @prefix prov: <http://www.w3.org/ns/prov#> .
3 @prefix dcterms: <http://purl.org/dc/terms/> .
4 @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
5
6 mmv:AlignProject a mmv:MappingProject ;
7   dcterms:title "Anatomy Ontology Alignment" ;
8   mmv:hasAnalysis mmv:AlignAnalysis ;
9   mmv:projectStartTime "2025-06-01T00:00:00Z"^^xsd:dateTime .
10
11 mmv:AlignAnalysis a mmv:AnalysisActivity ;
12   mmv:hasRequirement "Align mouse and human anatomy ontologies" ;
13   mmv:hasAssumption "Overlapping anatomical scope" ;
14   mmv:hasRiskOrIssue "Ambiguous labels across ontologies" .

```

Figure 2 MMV example for an ontology alignment use case

- Use Case 2: RML Uplift Mapping (Development Phase)

This use case (Figure 3) demonstrates how MMV represents development-phase metadata for an RML uplift mapping. A MappingProject is associated with a DevelopmentActivity describing the implementation of an uplift mapping from CSV data to RDF. The resulting MappingArtifact records the mapping type, mapping language, version, and reuse of an external mapping tool, illustrating how MMV documents how the mapping was implemented and which technologies were used.

```

1 @prefix mmv: <http://w3id.org/mmV/#> .
2 @prefix prov: <http://www.w3.org/ns/prov#> .
3 @prefix mqv: <https://alex-randles.github.io/MQV/#> .
4
5 mmv:RMLProject a mmv:MappingProject ;
6   mmv:hasDevelopment mmv:RMLDevelopment .
7
8 mmv:RMLDevelopment a mmv:DevelopmentActivity ;
9   prov:used mmv:RMLMapper ;
10  mmv:hasRequirement "Generate RDF from CSV source" ;
11  mmv:hasAssumption "Stable column identifiers" .
12
13 mmv:RMLMapping a mmv:MappingArtifact ;
14   mmv:mappingType "uplift mapping" ;
15   mmv:mappingLanguage "RML" ;
16   mmv:version "1.0" .

```

Figure 3 MMV example for an RML uplift mapping use case

- Use Case 3: Interlinking (Testing/Evaluation phase)

This use case (Figure 4) illustrates how MMV captures testing and evaluation metadata for an interlinking task. A MappingProject is linked to a TestingActivity that validates a generated linkset, while an EvaluationActivity records quality assessment results using explicit quality measurements. The example demonstrates how MMV represents validation outcomes and quantitative quality evidence, supporting informed reuse and comparison of interlinking artefacts.

¹ <https://sarahalzahranited.github.io/MMV1/>

```

1 @prefix mmv: <http://w3id.org/mmv/#> .
2 @prefix prov: <http://www.w3.org/ns/prov#> .
3 @prefix dqv: <http://www.w3.org/ns/dqv#> .
4 @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
5
6 mmv:LinkProject a mmv:MappingProject ;
7   mmv:hasTesting mmv:LinkTest .
8
9 mmv:LinkTest a mmv:TestingActivity ;
10  prov:used mmv:LinksetV1 .
11
12 mmv:LinksetV1 a mmv:MappingArtifact ;
13   mmv:mappingType "interlinking" ;
14   mmv:version "1.0" ;
15   dqv:hasQualityMeasurement mmv:QM1 .
16
17 mmv:QM1 a dqv:QualityMeasurement ;
18   dqv:computedOn mmv:LinksetV1 ;
19   dqv:isMeasurementOf mmv:PrecisionMetric ;
20   dqv:value "0.92"^^xsd:decimal .

```

Figure 4 MMV example for an Interlinking mapping use case

The complete MMV ontology, including its formal specification and documentation, is publicly available online¹.

IV. PLANNED EXPERT EVALUATION

A. Evaluation Goals

The planned expert evaluation aims to assess the conceptual clarity, coverage, and relevance of the Mapping Metadata Vocabulary (MMV) for documenting declarative mapping projects across their lifecycle. The evaluation focuses on whether MMV adequately represents metadata that domain experts consider important, including purpose, requirements, assumptions, validation, and maintenance information.

This study builds on an earlier survey of the Semantic Web community [5], which examined the relevance of different metadata elements for mapping-related tasks and informed the design of MMV. The current evaluation adopts a more focused perspective, examining whether MMV, as an ontology, captures these metadata requirements in a coherent and well-structured manner, while adhering to ontology engineering best practices such as conceptual clarity, appropriate abstraction, and reuse of existing vocabularies.

B. Evaluation Procedure

The expert evaluation is planned as a structured online survey targeting participants with experience in declarative mappings, ontology alignment, data integration, or knowledge graph construction. The primary objective of the evaluation is to assess the conceptual soundness, design quality, and coverage of the MMV ontology.

Participants will be asked to review selected parts of the MMV ontology documentation and respond to a survey combining three types of questions:

- Likert-scale questions assessing the clarity, relevance, and completeness of core ontology concepts and lifecycle phases.

- Targeted questions focusing on ontology modeling decisions, such as the level of abstraction and reuse of existing vocabularies.
- Open-ended questions allowing experts to identify missing concepts, ambiguities, or suggest improvements.

Domain experts from the Semantic Web community are invited to participate in the evaluation via an online survey available at the following expert Evaluation Survey². The study has received ethical approval from Trinity College Dublin, and participation is voluntary and anonymous

V. RELATED WORK

From an ontology engineering perspective, a recurring challenge is how to represent processes and their outputs in a way that supports traceability, comparison, and reuse. Prior work has emphasized the explicit modelling of activities, artefacts, and agents when ontologies are intended to support analysis, governance, or reuse, rather than purely conceptual representation. Provenance models such as PROV-O provide a general foundation for describing such process-oriented information, but do not prescribe domain-specific lifecycle structures or relationships tailored to declarative mapping projects [16].

Large-scale evaluation initiatives and shared tasks constitute an important source of mapping artefacts, as illustrated by the Ontology Alignment Evaluation Initiative (OAEI). These initiatives generate large collections of mapping artefacts accompanied by narrative descriptions of experimental settings, evaluation procedures, and assumptions. While this supports result comparability, the lack of formal, machine-interpretable representations of project context and lifecycle information limits traceability and reuse of individual artefacts beyond their original evaluation setting [9].

Recent community-driven initiatives such as FAIR-IMPACT have explicitly addressed the need for improved documentation of mappings and crosswalks to support Findability, Interoperability, and Reuse [17]. FAIR-IMPACT provides guidelines and a practical framework for creating, documenting, and sharing mappings, including recommendations on metadata elements, identifiers, formats, and governance practices. However, these contributions are primarily methodological and do not define a formal ontology for representing mapping projects, lifecycle activities, and their interrelationships in a machine-interpretable manner[17].

MMV complements these efforts by providing a lifecycle-based ontology that models mapping projects as first-class entities and explicitly links activities, artefacts, and evaluation outcomes within a coherent structure. This positioning allows MMV to support consistent documentation, reasoning, and reuse of declarative mapping projects across different mapping types and evaluation settings, while remaining compatible with existing guidelines and provenance standards.

² <https://forms.gle/Wg2KiZwgp84L6zBv8>

VI. TOWARD A PURPOSE-CENTRIC VOCABULARY

The development of MMV highlighted that explicitly capturing the purpose of an artefact is important not only for reuse decisions, but also for interpretation, evaluation, governance, and maintenance. In Semantic Web environments, artefacts such as mappings and datasets are frequently reused outside their original context. When purpose is not explicitly documented, this can lead to misinterpretation or inappropriate reuse.

While MMV captures purpose-related information within lifecycle activities, this work points to the value of modelling purpose as a first-class metadata element supported by a controlled vocabulary. A controlled vocabulary for purpose would promote consistency, comparability, and machine interpretability across artefacts, enabling clearer understanding of *why* an artefact was created and under which conditions it should be used.

As future work, we plan to explore a lightweight purpose vocabulary with predefined terms describing aspects such as project goal, application context, intended use, target audience, scope constraints, and reuse expectations. Such a vocabulary could complement lifecycle-based models like MMV and be applied more broadly across Semantic Web artefacts, beyond mapping projects alone.

VII. CONCLUSION AND FUTURE WORK

This paper introduced MMV, a lifecycle-based ontology for representing metadata about declarative mapping projects. MMV models mapping projects as first-class entities linked to lifecycle activities and resulting artefacts, enabling structured documentation of context, assumptions, implementation decisions, and evaluation outcomes across the full mapping lifecycle. By reusing established Semantic Web vocabularies, MMV integrates provenance and quality metadata into a unified, extensible framework.

The ontology has been illustrated using two representative use cases ontology alignment and RML uplift mapping demonstrating its applicability across different mapping types and lifecycle phases. These examples highlight how MMV can support traceability and reuse assessment without being tied to a specific mapping language or toolchain.

Future work will proceed in two main directions. First, an expert-based evaluation will be conducted to assess the conceptual clarity, coverage, and relevance of MMV from an ontology engineering perspective. Second, a reuse-oriented evaluation will investigate whether MMV-based metadata improves users' ability to assess the suitability of existing mappings for reuse in new contexts.

More broadly, this work contributes to ongoing efforts to improve the transparency, comparability, and sustainability of declarative mapping practices by treating metadata as a core component of mapping projects rather than an afterthought.

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REFERENCES

- [1] E. Iglesias, M.-E. Vidal, D. Collarana, and D. Chaves-Fraga, "Empowering the SDM-RDFizer tool for scaling up to complex knowledge graph creation pipelines1," *Semantic Web*, vol. 16, no. 2, p. SW-243580, 2025, doi: 10.3233/SW-243580.
- [2] A. Dimou, T. De Nies, R. Verborgh, E. Mannens, P. Mechant, and R. Van de Walle, "Automated metadata generation for Linked Data generation and publishing workflows," in *LDOW2016*, CEUR-WS. org, 2016, pp. 1–10.
- [3] F. Scharffe and D. Fensel, "Correspondence Patterns for Ontology Alignment," in *Knowledge Engineering: Practice and Patterns*, A. Gangemi and J. Euzenat, Eds., Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 83–92.
- [4] A. C. Junior, C. Debruyne, R. Brennan, and D. O'Sullivan, "FunUL: a method to incorporate functions into uplift mapping languages," in *Proceedings of the 18th International Conference on Information Integration and Web-Based Applications and Services*, in iiWAS '16. New York, NY, USA: Association for Computing Machinery, 2016, pp. 267–275. doi: 10.1145/3011141.3011152.
- [5] S. Alzahrani and D. O'Sullivan, "What metadata is needed for semantic and data mappings?," in *2023 IEEE 17th International Conference on Semantic Computing (ICSC)*, 2023, pp. 312–316. doi: 10.1109/ICSC56153.2023.00064.
- [6] S. Alzahrani and D. O'Sullivan, "MetaMap: Facilitating Mapping Reuse and Lifecycle Documentation," in *2024 IEEE 18th International Conference on Semantic Computing (ICSC)*, 2024, pp. 323–328. doi: 10.1109/ICSC59802.2024.00058.
- [7] S. Alzahrani and D. O'Sullivan, "MetaSEMAP: Usability Evaluation for Metadata Annotation Across the Mapping Lifecycle," in *First International Workshop on Users and Knowledge Graphs@ SEMANTiCS 2025*,
- [8] S. Alzahrani and D. O'Sullivan, "Guiding LLM Generated Mappings with Lifecycle-Based Metadata: An Early Evaluation," in *Joint Proceedings of Posters, Demos, Workshops, and Tutorials of the 21st International Conference on Semantic Systems co-located with 21st International Conference on Semantic Systems (SEMANTiCS 2025)*, Vienna, Austria, September 3-5, 2025, D. Chaves-Fraga, I. Heibi, D. Garijo, D. Collarana, A. A. Salatino, and S. Vahdati, Eds., in CEUR Workshop Proceedings, vol. 4064. CEUR-WS.org, 2025. [Online]. Available: <https://ceur-ws.org/Vol-4064/kgnesy-paper4.pdf>
- [9] "Ontology Alignment Evaluation Initiative::Home." Accessed: Dec. 15, 2025. [Online]. Available: <https://oaei.ontologymatching.org/>
- [10] "Ontology Alignment Evaluation Initiative::2025." Accessed: Dec. 15, 2025. [Online]. Available: <https://oaei.ontologymatching.org/2025/>
- [11] "Ontology Alignment Evaluation Initiative::Anatomy." Accessed: Dec. 15, 2025. [Online]. Available: <https://oaei.ontologymatching.org/2025/anatomy/index.html>
- [12] M. Poveda-Villalón, A. Fernández-Izquierdo, and R. García-Castro, "Linked Open Terms (LOT) Methodology (Version 1.0)." Ontology Engineering Group, Universidad Politécnica de Madrid, 2019. doi: 10.5281/zenodo.2539305.
- [13] "Data on the Web Best Practices: Data Quality Vocabulary." Accessed: Dec. 16, 2025. [Online]. Available: <https://www.w3.org/TR/vocab-dqv/>
- [14] "Mapping Quality Vocabulary." Accessed: Dec. 16, 2025. [Online]. Available: <https://alex-randles.github.io/MQV/>
- [15] S. Chávez-Feria, R. García Castro, and M. Poveda-Villalón, "Chowlk: from UML-Based Ontology Conceptualizations to OWL," 2022, pp. 338–352. doi: 10.1007/978-3-031-06981-9_20.
- [16] T. Lebo *et al.*, "PROV-O: The PROV Ontology," *W3C Recommendation*, 2013, [Online]. Available: <https://www.w3.org/TR/prov-o/>
- [17] M. Martinková, P. Winstanley, C. Debruyne, and others, "D2.2 – FAIR Digital Objects for mappings between schemas and ontologies," FAIR-IMPACT, 2024. doi: 10.5281/zenodo.15111167.